

Dyuthi (Dyu) Vijay

858-375-3240 | U.S. Citizen | dyuthi.vijay@gmail.com | linkedin.com/in/dyuthi-vijay | dyuthiv05.github.io

EDUCATION

University of California San Diego

MS in Data Science

San Diego, CA

Sept 2026 – Dec 2027

University of California Santa Cruz

BS Cognitive Science (AI Specialization), Minors in Statistics, Computer Science

Santa Cruz, CA

GPA: 3.9, Grad: June 2026

EXPERIENCE

AI Engineer

June '25 – Present

ASML

San Diego, CA

- Developed PyTorch models to predict part performance and failure modes at the design stage, using simulation feedback and hyperparameter tuning to improve accuracy 18%
- Built a gen AI agent framework with modular workflows and pipelines that automated 80% of documentation tasks for my team while maintaining internal standards
- Created a multi-agent blueprint for building end-to-end AI agents, including a cost framework for tokens, tool calls, and execution. One of the first use cases was a risk-assessment agent that reviewed 37 risk events, evaluated vendors, and recommended a vendor for our supply-chain team

Bioengineering Intern, Data Analytics

June '24 – Sept '24

Aquillius (BioTech Startup)

San Diego, CA

- Ran extensive statistical analysis on genomic datasets in Python and SQL, improving predictive model accuracy by 15% through targeted feature engineering such as aggregating mutation frequency by gene pathway.
- Built validation frameworks with stress tests and redundancy checks to model biological system reliability and reduce experimental error.

Data Analyst Intern, Financial Analytics

June '23 – Sept '23

Advanced Trustee Strategy Wealth Management

San Diego, CA

- Analyzed portfolios in Python and SQL, applying regression models to uncover risk exposures and investment opportunities that informed allocation decisions.
- Built interactive Power BI dashboards to visualize key financial metrics, cutting manual reporting time and giving stakeholders live insight for faster decisions.

PROJECTS

Agentic AI Blueprint & Evaluation Framework | *Python, LLMs, Agent Orchestration*

June '26 – Present

- Building a modular blueprint that assembles end to end AI agents from reusable components so teams can stand up task specific agents quickly.
- Designing an evaluation framework that scores agents on accuracy, reliability, and task completion to benchmark and improve agent designs.

Audio Keyword Classification | *Python*

Jan '24 – June '24

- Built a model that classifies keywords in audio clips by pairing signal processing with neural networks, reaching 87% accuracy on the Google Speech Commands dataset.

Slugbotics | *Hardware*

Jan '24 – June '26

- Built a robotic arm with servo motors and sensor feedback for steady, accurate movement, integrating five motor controllers for multi axis control.

RELEVANT COURSES

CSE 242: Machine Learning, CSE 240: Artificial Intelligence, STATS 132: Bayesian Statistics, STATS 131: Probability Theory, MATH 21: Linear Algebra, CSE 20: Python, CMPM 118: Neuromorphic Computing Group Research Lab

SKILLS

Languages: Python, SQL, C, MATLAB

Machine Learning & AI: PyTorch, TensorFlow, LLMs & Generative AI, Prompt Engineering, Neural Networks, Predictive Modeling, Model Evaluation, Feature Engineering

Data & Analytics: Statistical Modeling, Anomaly Detection, Data Validation & Quality, NumPy, Pandas, Tableau, Power BI, Plotly, Seaborn, Matplotlib

Foundations: Data Structures & Algorithms, APIs, ETL & Data Wrangling, Git